Comparing graph databases

There are several graph databases offerings https://db-engines.com/en/ranking/graph+dbms

See also https://www.marktechpost.com/2024/06/03/top-open-source-graph-databases/

Neo4j is the market leader

But there are many competitors:

- TigerGraph
- ArangoDB
- OrientDB
- Aerospike Graph
- etc etc

Some of these databases are multi-model: graph and documents

You are the CTO of a innovative, ground breaking, fully funded startup

Which graph database do you choose?

Pick one graph database from the websites above and compare it to Neo4j

Then we take 15mn to go around the room and discuss

Criterias

The list below is quite comprehensive.

Pick a subject, a specific topic and start investigating

Total Cost of Ownership

- Licensing costs (open-source vs. commercial)
- Cloud hosting fees (managed vs. self-hosted)
- Support costs and professional services
- Developer productivity (faster development = lower costs)

Scalability & Performance

Can it handle 10x, 100x your current data? What's the performance degradation curve? Can you add nodes without downtime?

Developer Experience

- Learning curve for your team
- Query language complexity (Cypher vs. SQL-like vs. proprietary)

Multi-Model Support

Reduce technical debt and complexity

- Document storage alongside graphs
- Key-value operations for caching
- Full-text search capabilities
- Geospatial features

Cloud-Native Readiness

- Managed cloud offerings (AWS, GCP, Azure)
- Kubernetes deployment options
- Backup/restore automation
- Monitoring and alerting capabilities
- Auto-scaling features

Vendor Lock-in Risk

Protect your future flexibility

- Open standards compliance
- Data export capabilities
- Query portability between systems
- Community vs. commercial ecosystem

Security & Compliance

- Authentication/authorization systems
- Data encryption (at rest and in transit)

- Audit logging capabilities
- Compliance certifications (SOC2, GDPR, etc.)

Ecosystem Integration

Leverage existing investments

- BI/Analytics tools integration (Tableau, PowerBI)
- Application frameworks support
- ETL/ELT pipelines compatibility
- Microservices architecture fit